

 **GROUP 7 SECTION 2**

RECOVERABLE ASSETS & INVENTORY RISK MANAGEMENT

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Case Study

PROBLEM

PPG Industries lacks real-time visibility into its inventory. Raw materials that exceed consumption levels accumulate as excess stock, risking obsolescence and costly write-downs. Without an integrated data system, inventory data remains siloed across procurement, production, and finance — making it impossible to detect Recoverable Assets, dead stock, or expired materials early enough to act.



Objective



1

To provide a layered architecture (Bronze, Silver, and Gold Layers) for end-to-end data engineering pipelines for inventory risk management.

2

To automatically identify and categorize inventory risk, including Recoverable Assets (RA), dead stock, expired supplies, and stockout hazards.

3

To increase data visibility and consistency by combining several dispersed data sources into a single data platform.

4

To put ETL/ELT procedures into place to guarantee that data is appropriately transformed, cleansed, and stored for analysis.

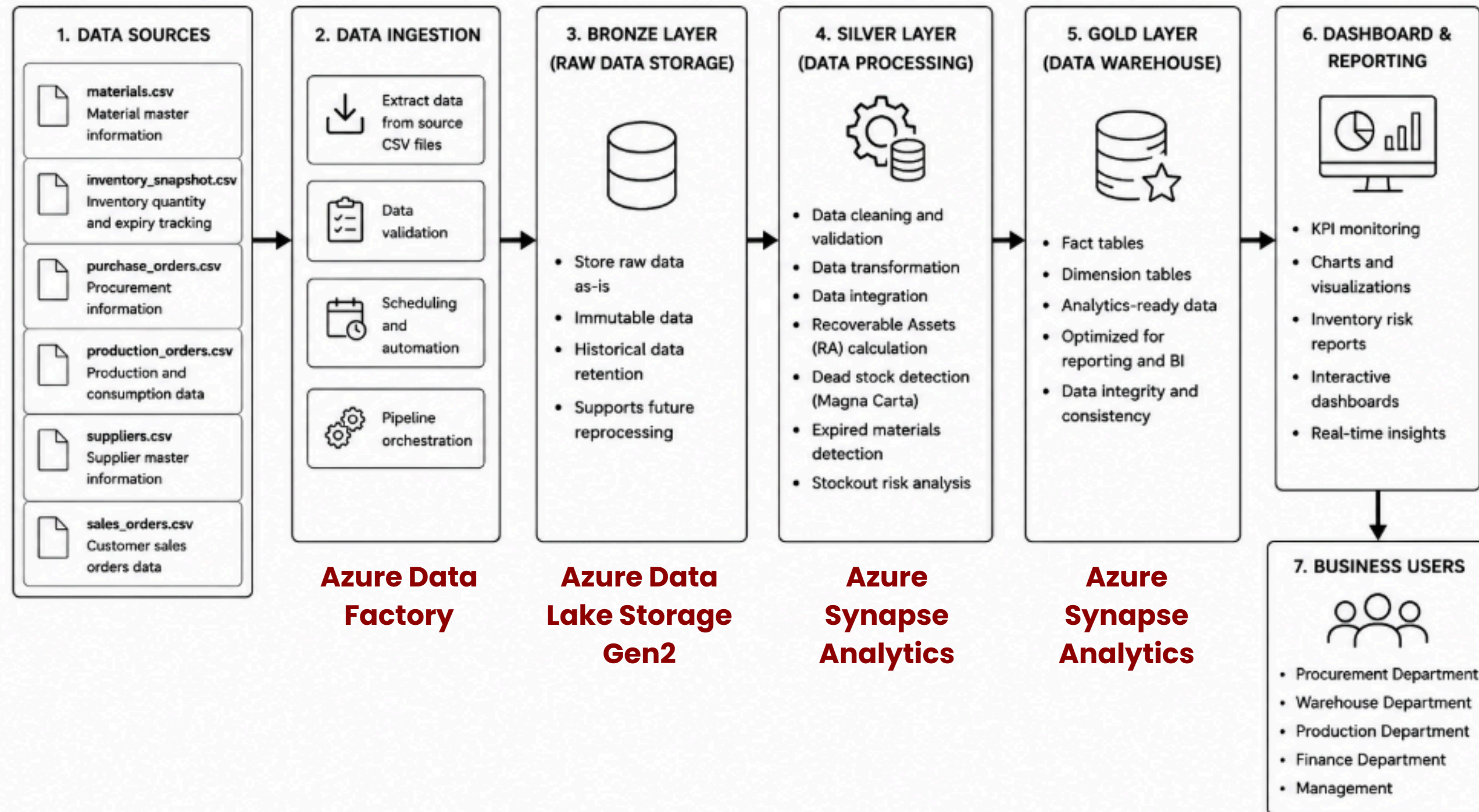
5

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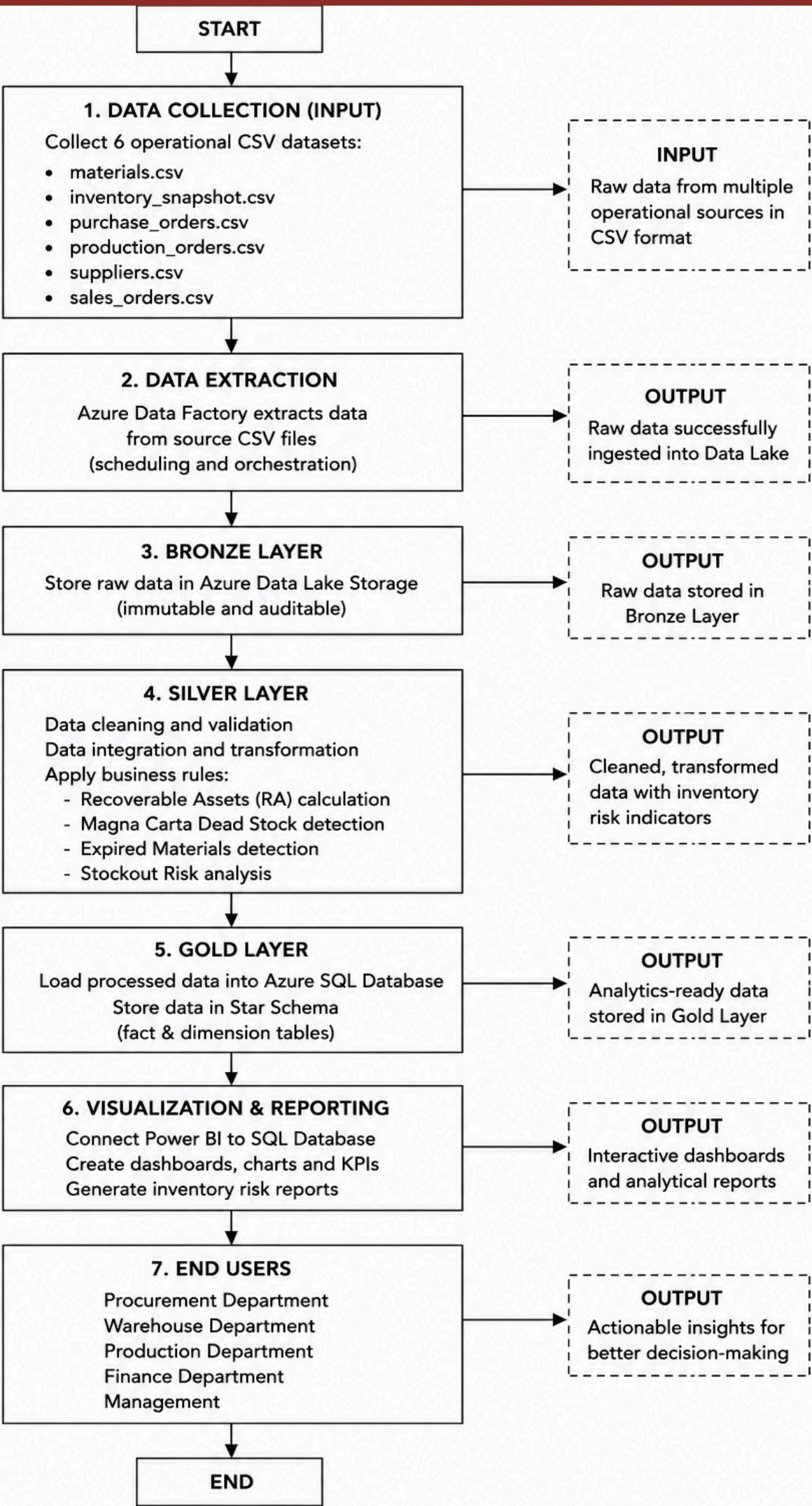
DATA ARCHITECTURE



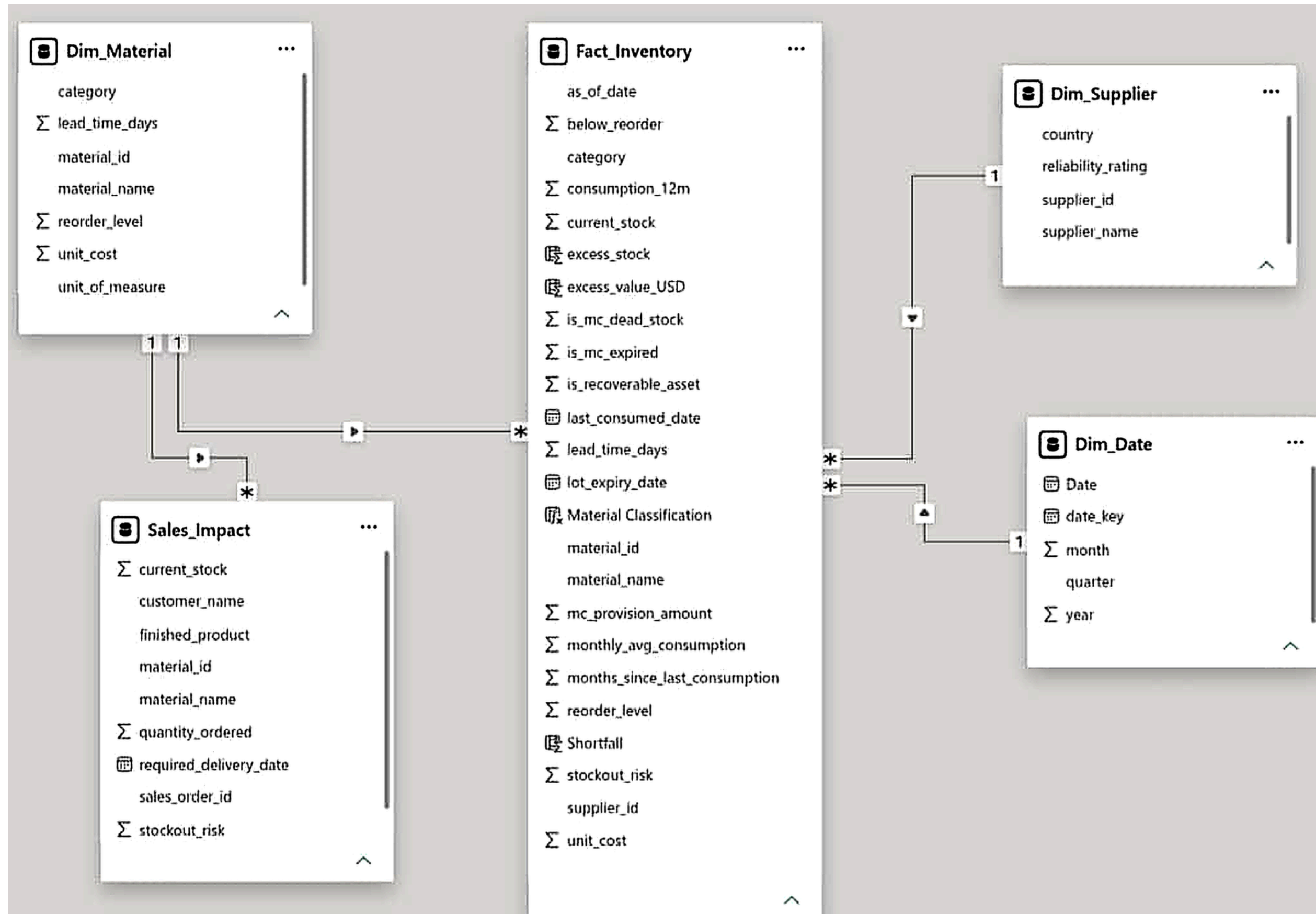
**Microsoft
Power BI**

SYSTEM IMPLEMENTATION FLOW

- Follows the Medallion Architecture (Bronze, Silver, Gold)
- Technology Used: Azure Data Lake Storage, Azure Data Factory., Azure Synapse Analytics, Power BI



GOLD LAYER SCHEMA



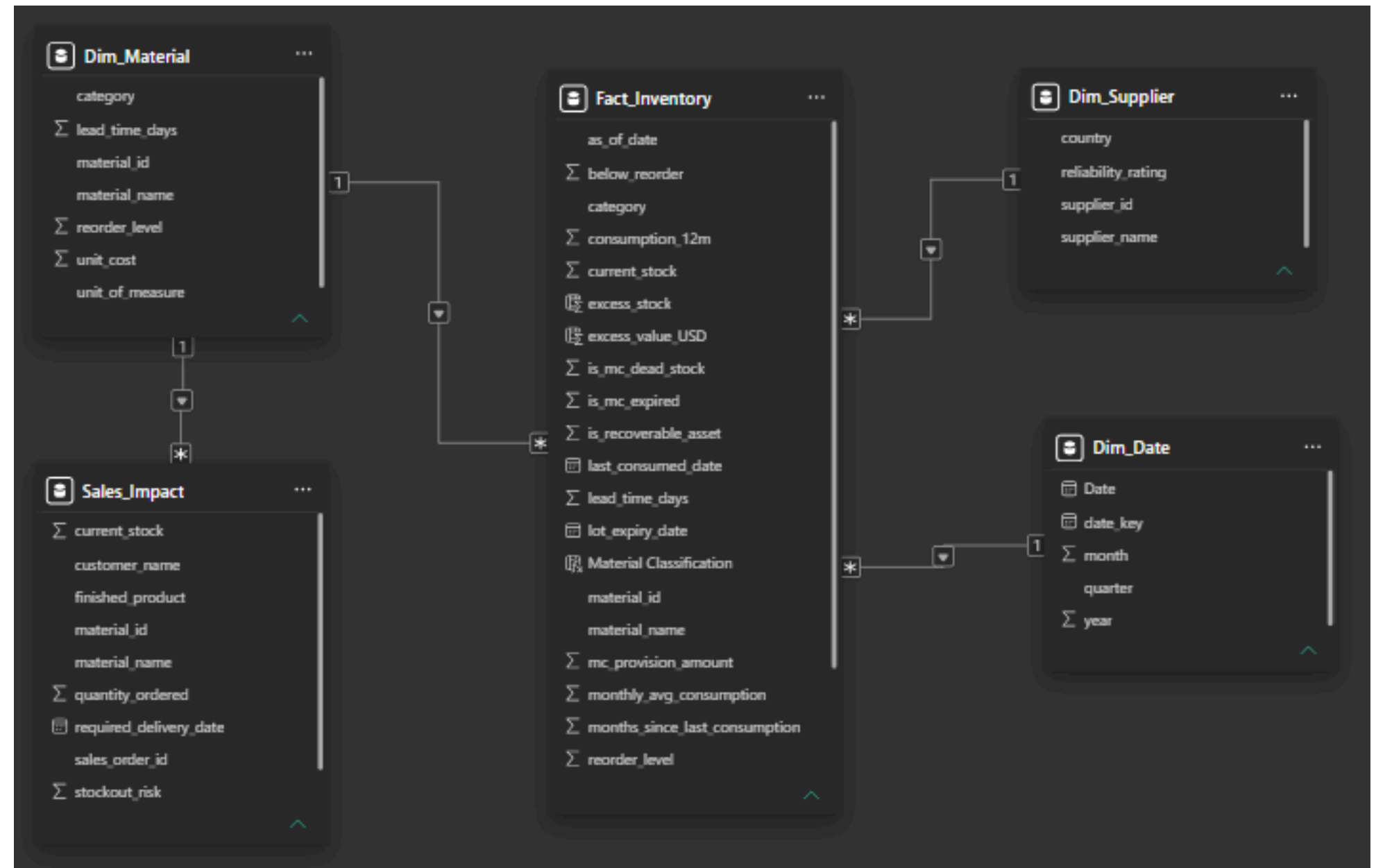
CURATED LAYER

01_Database_Setup

02_Create_External_Tables

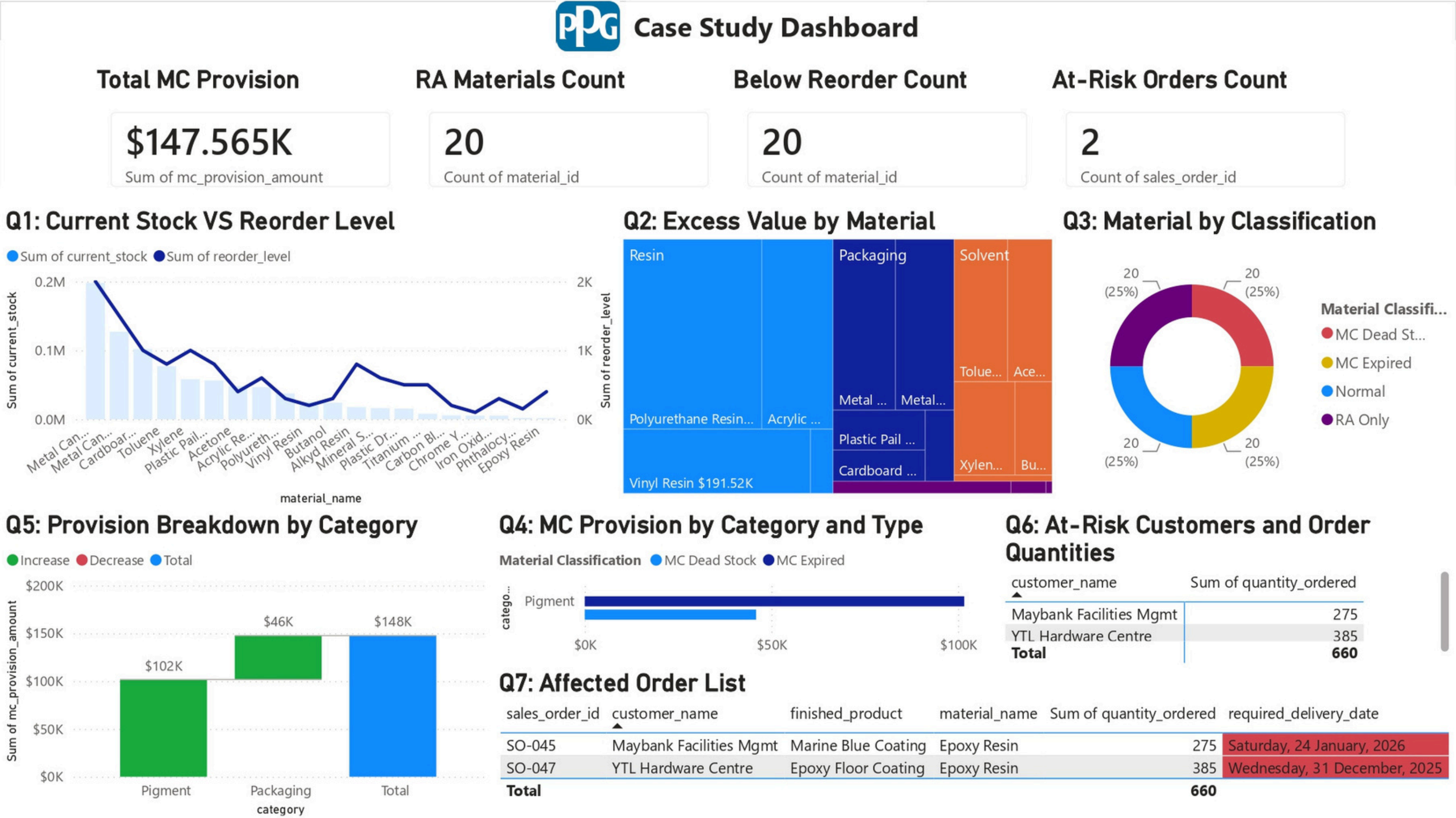
03_Business_Queries

The ppg_curated database serves as our analytics-ready Gold Layer, implementing a structured Star Schema to optimize query performance. It integrates business-logic-rich datasets—managed via automated CREATE EXTERNAL TABLE scripts—to provide a single, unified source of truth. This layer is to deliver high-speed, interactive results for all 7 core inventory risk business questions directly to the Power BI dashboard.



Star Schema

DASHBOARD



Conclusion

- **Successfully built an end-to-end Bronze–Silver–Gold Medallion Architecture pipeline on Microsoft Azure for PPG Industries inventory risk management**
- **Azure Data Factory and Synapse Analytics (Serverless SQL) automated data ingestion, cleaning, and transformation across all three layers with zero data loss**
- **All 7 business questions were answered accurately through the Gold Layer star schema, with all Black Box tests passing at every layer**
- **The 6-page Power BI dashboard provides real-time visibility into reorder alerts, Recoverable Assets, Magna Carta provision, and customer impact**
- **The system successfully eliminates data silos and quantifies financial exposure from excess and obsolete inventory, enabling data-driven decision-making across all departments**